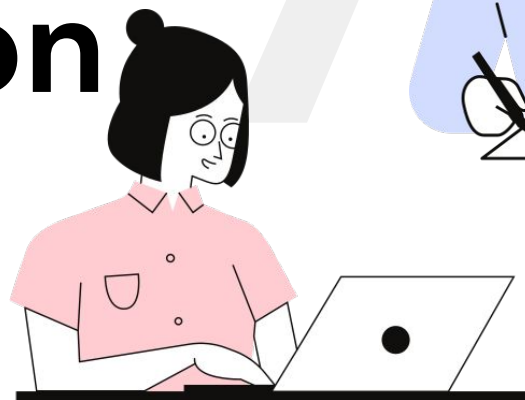


BRITISH INTERNATIONAL
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Information & Communication Technology



OBJECTIVES

- ❖ **Understand** the analysis stage, including system analysis methods, identifying inputs/outputs, and determining hardware/software needs.
- ❖ **Learn** the design stage elements, such as file structures, input/output formats, data capture forms, screen/report layouts, and validation routines.
- ❖ **Explore** the development and testing stage, covering test strategies, test plans, and different types of test data.
- ❖ **Examine** the implementation stage, including direct changeover, parallel running, pilot running, and phased implementation.
- ❖ **Understand** the documentation and evaluation stages, including technical/user documentation and comparing the new solution to the original system.

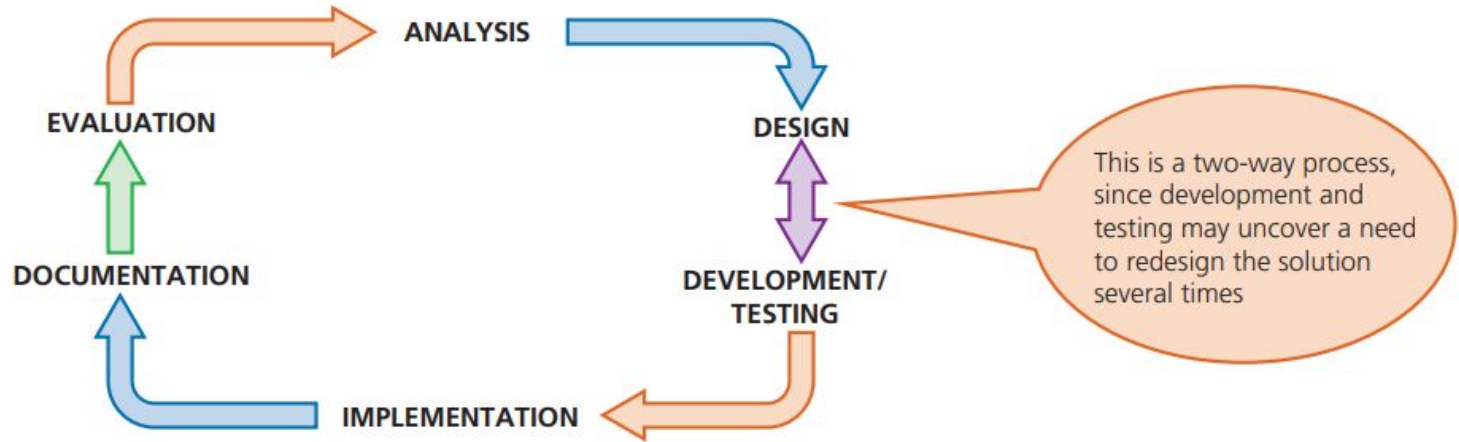
System Upgrade: Reasons & Process

A company may need to upgrade its computer system due to:

- ❖ **Obsolescence** – Existing equipment cannot be repaired.
- ❖ **Legal/tax changes** – New laws require system updates.
- ❖ **Better hardware available** – Enhances efficiency & reliability.
- ❖ **Business expansion** – Requires improved system capabilities.

- ❖ Oversees the upgrade process.
- ❖ Analyzes the current system.
- ❖ Suggests improvements, calculates costs, and reports advantages.
- ❖ Ensures smooth installation, evaluation, and modifications.

- ❖ **Analysis** – Understanding current system requirements.
- ❖ **Design** – Creating a new system plan.
- ❖ **Development & Testing** – Implementing & verifying system functions.
- ❖ **Implementation** – Installing and integrating the new system.
- ❖ **Evaluation & Maintenance** – Reviewing and making necessary updates.



- ❖ **Research current system** – Identify processes, inputs, outputs.
- ❖ **Identify problems** – Understand system weaknesses.
- ❖ **Determine user & information requirements.**
- ❖ **Recommend necessary hardware & software.**

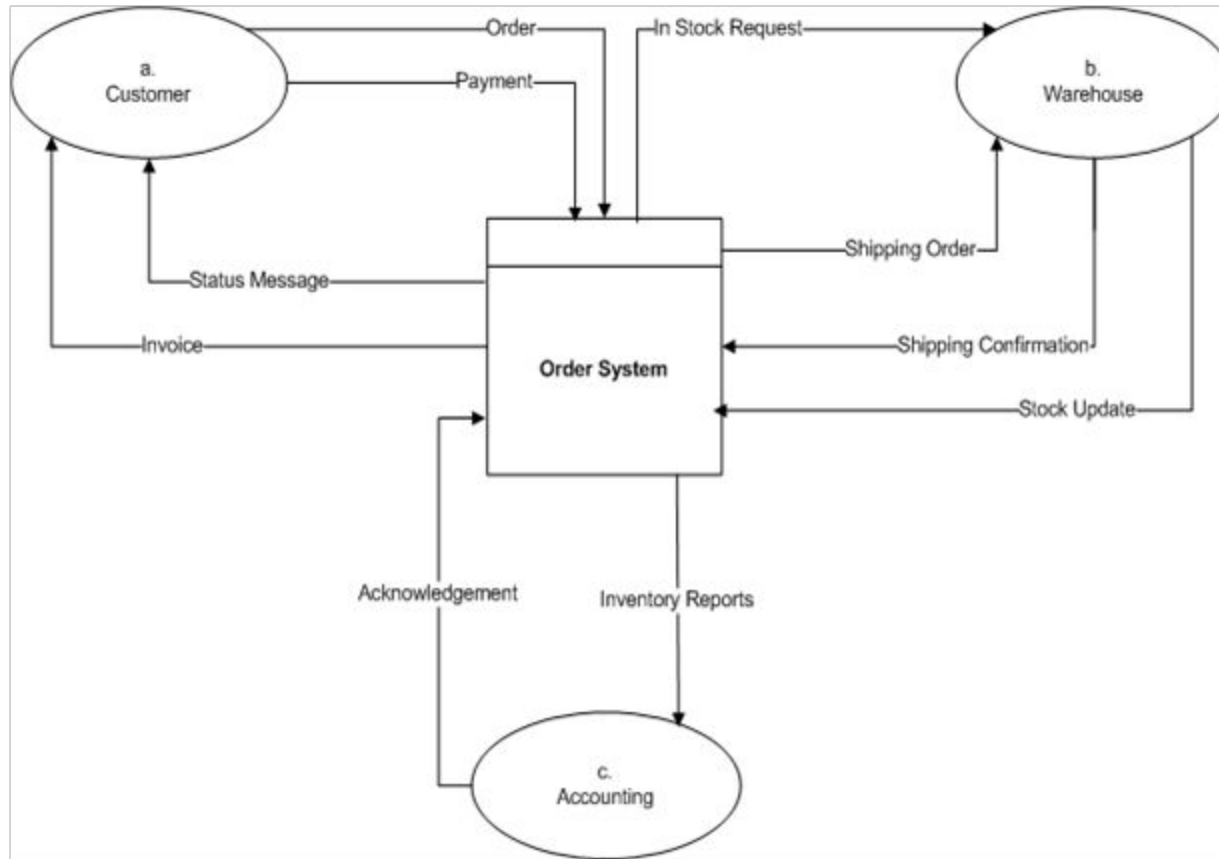
- ❖ Observation – Watching users interact with the system.
 - Reliable data, overall system view, cost-effective.
 - Users may alter behavior when observed (Hawthorne effect).
- ❖ Interviews – One-on-one discussions with users.
 - In-depth responses, flexible questioning.
 - Time-consuming, costly, possible bias in answers.
- ❖ Questionnaires – Distributed surveys to users.
 - Quick, cost-effective, allows anonymity.
 - Low response rate, rigid questions, no follow-ups.
- ❖ Existing Document Examination – Reviewing files, manuals, and reports.
 - Provides unique insights.
 - Time-consuming and expensive.

Identify:

- ❖ **Inputs** – Data entry methods.
- ❖ **Outputs** – System-generated results.
- ❖ **Processing** – System operations and transformations.
- ❖ **Problems** – Existing system inefficiencies.
- ❖ User & information requirements.

Flight Booking System (Data Flow Analysis)

- ❖ **Input:** Passenger details (name, passport, flight date).
- ❖ **Processing:** Checking flights, verifying data, assigning booking reference.
- ❖ **Output:** E-ticket, confirmation message.
- ❖ **Storage:** Database to keep booking records.



- ❖ Identifying Problems & Requirements
 - Analysts examine Data Flow Diagrams (DFD) to identify bottlenecks, errors in the current system, and determine user & information requirements.
- ❖ User Requirements
 - Bridge gap between business managers and system developers.
 - Written in natural language to ensure clarity.
 - Ensures the proposed system meets business needs, including accessibility.
- ❖ Information Requirements
 - What? (Data needed) & When? (Timing of data processing)
 - Used to create a functional requirements specification:
 - Defines customer, vendor, and employee interactions with the system.

Hardware Justification:

❖ Input Devices:

- Barcode readers – Efficient, reduces errors.
- Scanners – Convert paper documents to digital format.
- Touch screens – User-friendly for customer interactions.

❖ Output Devices:

- 3D printers – Cost-effective for prototyping.
- Large monitors – Needed for CAD design.
- Speakers – Assist employees with disabilities.

- ❖ Operating System – Chosen based on company needs.
- ❖ Application Software:
 - Off-the-shelf software – Cost-effective, minimal training needed.
 - Bespoke software – Tailored to needs but costly and time-intensive.
- ❖ Storage Considerations:
 - Size – Future-proofing required.
 - HDDs or SSDs – Based on access speed and durability.
 - Magnetic Tape – Suitable for large archives with minimal access needs.

Design Process Includes:

- ❖ **File & Data Structures** – Organizing data for efficiency.
- ❖ **Input Formats** – Designing user-friendly data entry methods.
- ❖ **Output Formats** – Structuring screen and report layouts.
- ❖ **Validation Routines** – Ensuring data integrity.

- ❖ Data is stored in records, each containing multiple fields.
- ❖ Primary Key – Uniquely identifies each record.

Example File Structure:

- ❖ **Fields:** Product Code, Year of Manufacture, Description, Price, Department.
- ❖ **Field Types:** Text, Numeric, Currency, Character.

- ❖ **Alphanumeric** – Mix of letters/numbers (e.g., A345FF).
- ❖ **Character** – Single letter/text (e.g., X).
- ❖ **Text** – String of characters (e.g., "example_text").
- ❖ **Boolean** – Logical values (Yes/No).
- ❖ **Numeric** – Used for calculations (Integer, Decimal, Currency).
- ❖ **Date/Time** – Stores date and time values.

Optimizing Field Lengths:

- Prevents wasted memory.
- Example: If max data is 5 characters, avoid setting field size to 50.

- ❖ Validation ensures data input meets predefined criteria but does not check accuracy.

Common Validation Checks:

- **Range Check** – Ensures input falls within a set range ($120 < \text{age} > 0$).
- **Type/Character Check** – Verifies data type (names w/o numbers).
- **Length Check** – Ensures input has a fixed number of characters (8-character password).
- **Format Check** – Ensures data follows a specific structure (date dd/mm/yyyy).
- **Presence Check** – Ensures required fields are not left empty (phone number).
- **Check Digit** – Extra digit added to detect errors in numeric sequences (preventing typos in ID numbers).

❖ Paper-Based Forms:

- Clear headings, text boxes, and tick boxes for easy data entry.
- Use of character boxes for structured input (e.g., names, phone numbers).
- Sufficient space for answers and readable fonts/colors.

❖ Computer-Based Forms:

- Text boxes, drop-down menus, and radio buttons for structured input.
- On-screen help & validation to guide users.
- Double-entry verification for key data like email.
- Control buttons (e.g., "Save", "Next", "Clear").
- Example (Car Sales Form Changes):
- Make of car → Drop-down menu for manufacturers.
- Date registered → Drop-down boxes for day, month, year.
- Price → Text box with validation.
- New/Used → Tickbox or radio button.

Screen Layouts:

- ❖ Correct field sizes for clarity.
- ❖ Clear instructions & labels for users.
- ❖ Full utilization of screen space to avoid blank areas.
- ❖ Appropriate fonts, colors, and contrast for readability.

Report Layouts:

- ❖ Includes headers, footers, and proper formatting.
- ❖ Data presented in tables for clarity (e.g., list of sales managers over 40).
- ❖ Ensures search criteria results are accurately displayed.

Need for Testing

- ❖ Ensures system functions correctly before going live.
- ❖ File Structure Finalization: determines data types, field lengths, key fields, and data linking.
- ❖ Validation & Verification: ensures accurate data transfer from paper-based to electronic systems.
- ❖ User Interface Testing: evaluates input/output devices and accessibility features (e.g., assistive technology).

- ❖ **Data Structures** – Ensures data is stored correctly.
- ❖ **File Structures** – Checks file functionality and data retrieval.
- ❖ **Input Formats** – Tests correct data entry.
- ❖ **Output Formats** – Verifies clarity and accuracy of outputs.
- ❖ **Validation Routines** – Ensures invalid data is rejected.

- ❖ Modular Testing:
 - Software is broken into smaller modules and tested separately.
 - Each module is modified if errors occur and re-tested.
- ❖ System Testing:
 - Once all modules are combined, the entire system is tested to detect compatibility issues.

- ❖ A test plan includes:
 - List of tests, expected outcomes, and comparison with actual results.
 - Types of test data:
 - **Normal Data** – Expected inputs within valid range (e.g., month: 1-12).
 - **Extreme Data** – Boundary values (e.g., month: 1 or 12).
 - **Abnormal Data** – Invalid inputs causing errors (e.g., letters in a numeric field).

- ❖ Live Data Testing:
 - Uses real data to compare new vs. existing system results.
 - If discrepancies arise, system modifications are required.

- ❖ Valid Range: 21°C to 50°C
- ❖ Expected Behavior:
 - **Accepted:** Normal (35, 45), Extreme (21, 50).
 - **Rejected:** Abnormal (e.g., "fifty", 20, -5, 100.5).

If actual results differ, system modifications are needed.

Implementation Stages

- ❖ **Data Transfer:** Key in data, scan documents, transfer paper/electronic files.
- ❖ **Staff Training:** Essential to ensure smooth transition

Changeover Methods

- ❖ Direct Changeover.
- ❖ Parallel Running
- ❖ Pilot Implementation
- ❖ Phased Implementation

Direct Changeover

- ❖ Old system stops, new system starts immediately.
- ❖ Immediate benefits, lower cost ✓
- ❖ Risky—failure can be disastrous ✗

Parallel Running

- ❖ Old and new systems run together for a period.
- ❖ Safer—old system acts as backup ✓
- ❖ Costly and time-consuming (data entered twice) ✗

Pilot Implementation

- ❖ New system tested in one department before full rollout.
- ❖ Limits failure impact, cost-effective training ✓
- ❖ Affected department may face challenges ✗

Phased Implementation

- ❖ System introduced in parts over time.
- ❖ Allows gradual adaptation, minimizes risk ✓
- ❖ Slow process, more expensive than direct changeover ✗

Technical Documentation (for programmers/analysts)

- ❖ **System details:** Code, programming language, flowcharts, algorithms.
- ❖ **Hardware & software requirements:** Minimum specs needed.
- ❖ **File structures:** Database format, validation rules.
- ❖ **Known bugs & error messages:** Troubleshooting support.

User Documentation (for end-users)

- ❖ **System operation:** Installation, saving, searching, sorting, printing.
- ❖ **User interface:** Input formats, screen layouts, print layouts.
- ❖ **Troubleshooting:** Error messages, FAQs, help guides.
- ❖ **Training materials:** Tutorials, login/logout procedures.

Assessing System Performance

- ❖ Compare final system with initial requirements.
- ❖ Identify limitations & possible improvements.
- ❖ Gather user feedback (interviews, questionnaires).
- ❖ Compare new vs. old system performance:
 - Task completion time
 - Ease of use
 - Business efficiency improvements

Potential Updates

- ❖ **Hardware updates:** New technology, business needs, user feedback.
- ❖ **Software updates:** User feedback, company changes, legal requirements.